MIL-T-60227A(MU)
10 May 1974
SUPERSEDING
MIL-T-60227(MU)
27 May, 1965

MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M7 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated, M7 Assembly, loaded with Cartridge Impulse, M67A1 for use in conjunction with other cartridge actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-P-19834	- Plates, Identification, Metal Foil Adhesive Backed

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MS-28741	- Hose Assembly - Detachable End Fitting, Medium Pressure

DRAWINGS

Armament Command

F8593277	- Thruster, Cartridge Actuated, M7 Assembly

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:21Z
Check the source to verify that this is the current version before use.

2.1 (Cont'd)

C8796479 - Box, Packing, Ammunition for Thruster, Cartridge Actuated, M7 with Cartridge Impulse, M67A1

.LLATION
ns
ry

C PREPARATION FOR INSTA
321 Safety precauti
324 Formation

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:21Z
Check the source to verify that this is the current version before use.

3.5 Functioning requirements.

3.5.1 The thruster shall not fail to fire.

3.5.2 The thruster shall function without mechanical failure (See 6.1).

3.5.3 The thruster fired under load shall perform as follows: (See 6.4)

3.5.3.1 Load.-The thruster shall propel an 11.6 pound weight, horizontally throughout its stroke of 2 2 1/2 inches ± 1/16 inch.

3.5.3.2 Thrust.-The thrust developed at 1/2 inch of stroke shall be 500 pounds, minimum. The final thrust at the end of stroke shall be 260 lbs., minimum.

3.5.3.3 Ignition delay.-The ignition delay time shall not exceed 65 milliseconds. (See 6.5)

3.5.3.4 Stroke time.-The time to complete the thruster stroke shall not exceed 70 milleseconds.

3.5.4 Temperature.-The thruster shall comply with all the ballistic requirements throughout the temperature range of -65°F to +165°F.

3.5.5 Initiation.-The thruster shall be fired by a source delivering an output equal to that of an M3A2 Initiator at the end of 15 feet of aircraft hose, size 4, MS28741, at 70° ± 5°F.

3.6 Workmanship.-The requirements for workmanship are as specified on the applicable drawing, referenced specifications and the following:

3.6.1 Metal defects.-All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.6.2 Burr.-No part shall have a burr which might interfere with the assembly or function of the item which might be injurious to personnel handling the item.

3.6.3 Foreign matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification and drawing where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:21Z
Check the source to verify that this is the current version before use.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.-A lot size shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 to 300	100% Examination	-	-	-	-
301 to 1300	110	1	2	2	3
1301 to 3200	150	2	3	3	4
3201 to 8000	225	3	4	4	5
8001 to 22000	300	4	5	5	6

4.2.2.1 Cartridge, Impulse, M67A1 Assembly, Drawing D8597730, covering a detail of Drawing F8593277.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. "O" ring missing	Visual
4. Sealing disc missing, damaged or improperly secured	Visual
5. Marking missing, incorrect or illegible	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:21Z
Check the source to verify that this is the current version before use.

4.2.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. "O" ring split, cut or cracked	Visual
102. Foreign matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Head, Guide and Firing Pin, Subassembly, consisting of Head 8593279, Guide 8593282, Firing Pin 8593864, Shear Pin 8593284 and O-Ring MS-28775 and MS-29513, covering a detail of Drawing F8593277.

CRITICAL	
1. Firing pin hole obstructed	Visual
MAJOR	
101. Thread damaged or not full	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Body with Cartridge, consisting of Body 8593280, End Sleeve 8593281, Piston 8593285, Collar 8593286, Piston Locking Spring 8593237, Key 8593288, M67A1 Cartridge 8597730, O-Ring MS-28775 and MS-29513-15, covering a detail of Drawing F8593277.

CRITICAL	
1. Mixed cartridges	Visual
MAJOR	
101. Any part missing, except as otherwise classified	Visual
102. Thread damaged or not full	Visual
103. Burr	Visual
104. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:21Z
Check the source to verify that this is the current version before use.

4.2.2.3 (Cont'd)

Defect	Method of Inspection
MINOR	
None defined	

4.2.2.4 Assembly, less Seal and Shipping Plugs, Drawing F8593277, less B8593299, A10535500 and B8594697.

CRITICAL	
1. Shear pin not engaged in firing pin	Gage
2. Guide not in contact with cartridge	Gage
3. Foreign matter in gas port	Visual
MAJOR	
101. Total length	Gage
102. Thread damaged	Visual
MINOR	
201. Set screw above flush	Visual
202. Set screw not filled with barrier material	Visual

4.2.2.5 Assembly, Drawing F8593277.

CRITICAL	
1. Mixed thrusters	Visual
MAJOR	
101. End sleeve not free to rotate 360 degrees clockwise	Manual
102. Marking, missing, incorrect or illegible	Visual
103. Foreign matter	Visual
MINOR	
201. Total length	Measure
202. Location of labels, incorrect	Visual
203. Seal missing or crimp, improper	Visual
204. Shipping plugs missing	Visual
205. Protective finish inadequate or defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:21Z
Check the source to verify that this is the current version before use.

4.2.2.6 Packing in Box, Fiberboard (Inner), Drawing C8796480.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Box broken or damaged	Visual
203. Number of units contained incorrect	Visual
204. Marking missing, incorrect or illegible	Visual
205. Tape incomplete or badly	Visual

4.2.2.7 Box, Packing Ammunition (Level A), Drawing C8796479.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	Manual
201. Contents loose	Manual/Visual
202. Strapping missing, loose, broken or improperly assembled	Visual
203. Contents of box exposed	Visual
204. Box damaged	Visual
205. Marking misleading and unidentifiable	Visual
206. Number of units incorrect	

4.2.2.8 Carton, Alternative Packing, Drawing C9234882.

CRITICAL	
None defined	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:21Z
Check the source to verify that this is the current version before use.

4.2.2.8 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Carton broken or damaged	Visual
203. Number of units incorrect	Visual
204. Marking missing, incorrect or illegible	Visual
205. Tape incomplete	Visual

4.2.3 Inspection testing.-The following tests shall be performed in accordance with 4.3:

4.2.3.1 Breakaway torque.

4.2.3.1.1 Head and body.-Each head and body assembly shall be subjected to a breakaway torque. Failure of any assembly to comply with the requirements of 3.2 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism.-Each Thruster shall be tested for excessive residual magnetism. Failure of any thruster to comply with the requirements of 3.4 shall be cause for rejection of the thruster.

4.2.3.3 Functional tests.-The following tests shall be performed by the Government:

4.2.3.3.1 Ballistics.-Fourteen units shall be tested to determine compliance with the requirements of 3.5.1 through 3.5.5. The unit shall be temperature conditioned as specified in Table IV and tested in accordance with 4.3.4. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

TABLE III

Temperature	Sample Size	Method
-65° to -70°F	7	Load
+160° to +165°F	7	Load

4.2.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-8593277.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:21Z
Check the source to verify that this is the current version before use.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.

4.3.1.1 Head and body.-A wrench, set at the torque requirement specified in Table I, shall be used in conjunction with a fixture to determine the adequacy of the joint. The breakaway torque shall be applied to the head after insertion of cartridge and set screws. An air vise or other holding fixture shall be used to secure the body.

4.3.2 Residual magnetism.-The Thruster shall be tested for excessive residual magnetism, in an area free of local magnetic effects, by placing the thruster 6 inches from and at the same heitht as the compass, in a north-south horizontal position with compass set in an east-west heading as shown in Figure 1. The thruster shall be moved slowly pass the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster around its longitudinal axis.

4.3.3 Functional.

4.3.3.1 Ballistics.-The test samples shall be rigidly supported so that the entire thrust is borne by the thruster mounting points. The sample shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperature.

4.3.3.1.1 Temperature conditioning.-A simulated thruster and cartridge with a temperature sensitive element, placed in the dummy cartridge, shall be stored in a temperature conditioning chamber at the same time as the samples. The temperature conditioning time shall be a minimum of one hour longer than the time necessary for the temperature of the simulated cartridge to reach the required test temperature. The total conditioning time including reconditioning at plus 165°F shall not exceed 18 hours.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be as specified by the procuring activity: Level A - in accordance with Drawings C8796479 and C8796480; Level B - in accordance with Drawings C9234882 and C8796480.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breaking of a part the occurrence of which is other than a design function.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:21Z
Check the source to verify that this is the current version before use.

6.2 Load.-The expected performance of the thruster, under condition simulating those existing in aircraft, shall be referred to as load condition in this specification.

6.3 Ignition delay is defined as the time lpased between the firing of the cartridge and the beginning of the continuous pressure rise on the body of the thruster.

6.4 Explosive Hazard Classification Information:

Explosive Hazard Classification (Quantity Distances): Class 1
Storage Compatibility: Group B, E or N
Department of Transportation (DOT) Hazard - Class C
DOT Marking - Explosive Power Device - Class C
Handle Carefully, Keep Fire Away

6.5 Hazard notice.-The thruster described herein and certain of its components are flammable and/or explosive, consequentlv present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

6.6 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A577

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-60227A(MU) Thruster, Cartridge Actuated, M7 Assembly

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO
	DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:21Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Philadelphia, Penna. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:21Z
Check the source to verify that this is the current version before use.